

Subject 1: Computer Fundamentals & Digital Logic

◆ Unit 1 – Fundamentals of Computers

Most Important Topics:

- Definition, Characteristics, and Types of Computers
- Block diagram / basic organization of a computer
- Input & Output devices (Peripheral devices examples)
- Memory hierarchy: Primary vs Secondary memory
- Types of Software (System, Application, Utility)
- Language Translators – Compiler, Interpreter, Assembler

Likely Questions:

1. Explain the basic organization of a digital computer with a neat diagram.
 2. Differentiate between system software and application software.
 3. Write short notes on language translators.
 4. Explain types of computer languages with examples.
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◆ Unit 2 – Number Systems

Most Important Topics:

- Conversion between Decimal, Binary, Octal, Hexadecimal
- 1's and 2's complement representation and subtraction
- Binary addition and subtraction
- Binary Coded Decimal (BCD)
- Gray Code and Excess-3 conversions

Likely Questions:

1. Convert $(101101)_2$ to decimal, octal, and hexadecimal.
 2. Perform subtraction using 2's complement.
 3. Explain Binary Coded Decimal with example.
 4. What is Gray code? Convert $(1010)_2$ to Gray code.
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◆ Unit 3 – Boolean Algebra

Most Important Topics:

- Boolean Laws & De Morgan's Theorems
- Duality Principle
- Canonical forms: SOP and POS
- Karnaugh Map (up to 4 variables)
- Basic Gates, Derived Gates, Universal Gates

Likely Questions:

1. State and prove De Morgan's Theorems.
 2. Simplify using Boolean algebra: $AB + A'B'$.
 3. Simplify using K-map (4 variables).
 4. What are universal gates? Why NAND/NOR are called universal gates?
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◆ Unit 4 – Logic Circuits

Most Important Topics:

- Half adder and Full adder/subtractor
- Flip-Flops (SR, JK, T, Master-Slave JK)
- Encoders and Decoders
- Multiplexers and Demultiplexers
- Counters (Ripple counter), Shift Registers

Likely Questions:

1. Explain the working of Half Adder and Full Adder with truth table and logic diagram.
2. Write short note on JK Flip-Flop.
3. Explain 4-to-1 multiplexer with diagram.
4. What is a counter? Explain 3-bit ripple counter.

Subject 2: Discrete Mathematics & Statistics

◆ Unit 1 – Logic and Set Theory

Most Important Topics:

- Logical connectives (AND, OR, NOT, Implication, Biconditional)
- Truth Tables
- Tautology and Contradiction
- Set operations (Union, Intersection, Complement)
- Venn Diagrams

Likely Questions:

1. Construct truth table for $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$.
 2. Define tautology and contradiction with example.
 3. Explain various set operations with examples.
 4. Represent $(A \cap B') \cup C$ in Venn diagram.
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◆ Unit 2 – Relations and Functions

Most Important Topics:

- Cartesian Product
- Properties of Relations (Reflexive, Symmetric, Transitive)
- Equivalence Relations and Partitions
- One-to-One, Onto, Inverse, Composite Functions

Likely Questions:

1. Define and give example of equivalence relation.
 2. Check whether the relation $R = \{(1,1), (2,2), (1,2)\}$ is symmetric or not.
 3. Define one-to-one and onto functions with examples.
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◆ Unit 3 – Statistics Basics

Most Important Topics:

- Frequency Distribution

- Arithmetic Mean (Grouped & Ungrouped)
- Median and Mode
- Measures of Central Tendency

Likely Questions:

1. Find arithmetic mean for given data.
 2. Calculate median for ungrouped data.
 3. Define mode and calculate mode for given discrete series.
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◆ ***Unit 4 – Statistical Analysis***

Most Important Topics:

- Standard Deviation (Actual Mean & Step Deviation)
- Coefficient of Variation
- Correlation (Types, Rank correlation)
- Simple Linear Regression

Likely Questions:

1. Calculate standard deviation using step deviation method.
2. Define coefficient of variation and its significance.
3. Explain types of correlation.
4. Define regression and properties of regression coefficients.

Subject 3: C++ Programming

◆ Unit 1 – Introduction to Programming and C++

Most Important Topics:

- OOP vs POP (Object-oriented vs Procedure-oriented)
- Structure of a C++ program
- Tokens: Keywords, Identifiers, Variables, Constants
- Data Types, Operators & Expressions
- Type conversion and Operator precedence

Likely Questions:

1. Explain the basic structure of a C++ program with example.
 2. Differentiate between POP and OOP.
 3. Define variable and constant in C++.
 4. Write short notes on data types and type conversion.
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◆ Unit 2 – Control & Modular Programming

Most Important Topics:

- If, If-Else, Nested If, Switch
- Loops: For, While, Do-While
- Functions: Definition, Declaration, Call by Value & Reference
- Inline Functions, Function Overloading
- Recursion

Likely Questions:

1. Explain different types of control statements with examples.
 2. Write a recursive function to find factorial.
 3. What is function overloading? Give example.
 4. Differentiate between call by value and call by reference.
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◆ Unit 3 – Derived Data Types and I/O Operations

Most Important Topics:

- Arrays (1D, 2D)
- Strings and String Manipulation
- Pointers and Pointer Arithmetic
- C++ Stream Classes (cin, cout)
- Formatted and Unformatted I/O

Likely Questions:

1. Write a program to add two matrices.
 2. Explain pointer arithmetic with example.
 3. What is a stream in C++?
 4. Write a program to find length of a string using pointers.
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◆ Unit 4 – OOP Concepts**Most Important Topics:**

- Class and Object
- Constructors & Destructors
- Friend Function
- Operator Overloading
- Inheritance (Types)
- Virtual Functions & Abstract Classes
- File Handling (Opening, Reading, Writing, Closing)

Likely Questions:

1. Define class and object with example program.
2. Explain types of constructors in C++.
3. Write a program to overload '+' operator.
4. What is inheritance? Explain its types.
5. Explain file handling operations in C++.

